

Order Service API — Test Report

Manual endpoint verification via curl against http://localhost:8080 | Run on: macOS (Nandans-MacBook-Air), working directory Assignment5

#	Endpoint	Method	Status	Result
1	/orders	GET	200	PASS
2	/orders	POST	201	PASS
3	/orders	GET	200	PASS
4	/orders/1	GET	200	PASS
5	/orders/1	GET	200	PASS
6	/orders/1/cancellation	POST	200	PASS
7	/orders/1	DELETE	204	PASS
8	/orders/999	GET	404	PASS
9	/orders	GET	401	PASS
10	/orders/1	GET	304	PASS
11	/orders	POST	200	PASS
12	/orders/1	PUT	412	PASS

Sequencing note: tests 1–9 were executed in order against a single application run. Order 1 is deleted in test 7, so tests 10–12 were executed after restarting the service and re-creating order 1 with the same request as test 2.

1. List Orders — Initial State (Empty)

Method & Path	GET /orders
Request Headers	Authorization: Bearer demo
Status	200
Response Body	[]

Notes: Returns an empty JSON array before any orders have been created. Confirms the endpoint is reachable and authenticated correctly.

2. Create Order

Method & Path	POST /orders
Request Headers	Authorization: Bearer demo Content-Type: application/json Idempotency-Key: abc123
Request Body	{ "customerName": "Nandan", "totalAmount": 500 }

Status	201
Response Headers	Location: /orders/1 X-RateLimit-Limit: 100 X-RateLimit-Remaining: 99
Response Body	<pre>{ "orderId": 1, "customerName": "Nandan", "totalAmount": 500, "status": "CREATED" }</pre>

Notes: Order created successfully with HTTP 201. Response includes a Location header pointing to the new resource (/orders/1) and rate-limit headers. Idempotency-Key was supplied to guard against duplicate submissions.

3. List Orders — After Creation

Method & Path	GET /orders
Request Headers	Authorization: Bearer demo
Status	200
Response Body	<pre>[{ "orderId": 1, "customerName": "Nandan", "totalAmount": 500, "status": "CREATED", "createdAt": "2026-09-19T05:28:55.262595Z" }]</pre>

Notes: The previously created order is now present in the collection, including a server-generated createdAt timestamp.

4. Get Order by ID

Method & Path	GET /orders/1
Request Headers	Authorization: Bearer demo
Status	200
Response Headers	ETag: "1453530583" Cache-Control: max-age=60
Response Body	<pre>{ "orderId": 1, "customerName": "Nandan", "totalAmount": 500, "status": "CREATED" }</pre>

Notes: Single-resource fetch succeeds and returns caching metadata (ETag, Cache-Control: max-age=60) suitable for conditional requests.

5. Get Order with Non-Matching If-None-Match

Method & Path	GET /orders/1
Request Headers	Authorization: Bearer demo If-None-Match: "12345"
Status	200
Response Headers	ETag: "483808708" Cache-Control: max-age=60
Response Body	<pre>{ "orderId": 1, "customerName": "Nandan", "totalAmount": 500, "status": "CREATED" }</pre>

Notes: Since the supplied If-None-Match value does not match the current ETag, the server returns 200 with the full body and a fresh ETag, rather than 304 Not Modified.

6. Cancel Order

Method & Path	POST /orders/1/cancellation
Request Headers	Authorization: Bearer demo
Status	200
Response Body	<pre>{ "orderId": 1, "customerName": "Nandan", "totalAmount": 500, "status": "CANCELLED" }</pre>

Notes: Order status transitions from CREATED to CANCELLED via a dedicated cancellation sub-resource endpoint.

7. Delete Order

Method & Path	DELETE /orders/1
Request Headers	Authorization: Bearer demo
Status	204
Response Body	<i>(empty)</i>

Notes: Deletion succeeds with 204 No Content and an empty response body, as expected for a successful DELETE.

8. Get Non-Existent Order

Method & Path	GET /orders/999
Request Headers	Authorization: Bearer demo
Status	404

Response Body	<pre>{ "title": "Not Found", "status": 404, "detail": "Order not found" }</pre>
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Notes: Order 999 does not exist (and order 1 was deleted in test 7). The API returns 404 Not Found with a problem-detail style body, which is the correct status for a missing resource.

9. List Orders — Missing Authorization

Method & Path	GET /orders
Request Headers	(no Authorization header)
Status	401
Response Body	<pre>{ "timestamp": "2026-09-19T05:30:11.488+00:00", "status": 401, "error": "Unauthorized", "path": "/orders" }</pre>

Notes: Request without a Bearer token is correctly rejected with 401 Unauthorized and Spring Boot's default error payload shape.

10. Conditional GET — 304 Not Modified

Method & Path	GET /orders/1
Request Headers	Authorization: Bearer demo If-None-Match: "1453530583"
Status	304
Response Body	(empty)

Notes: The client already possesses the current representation identified by the ETag. The server therefore returns 304 Not Modified and omits the response body, saving bandwidth and reducing latency.

11. Create Order Again With Same Idempotency-Key

Method & Path	POST /orders
Request Headers	Authorization: Bearer demo Content-Type: application/json Idempotency-Key: abc123
Request Body	<pre>{ "customerName": "Nandan", "totalAmount": 500 }</pre>
Status	200

Response Body	<pre>{ "orderId": 1, "customerName": "Nandan", "totalAmount": 500, "status": "CREATED" }</pre>
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Notes: The same Idempotency-Key was reused. Instead of creating a duplicate order, the server returned the existing resource with 200 OK rather than 201 Created. This prevents accidental duplicate submissions caused by client retries or double-clicks.

12. Conditional Write Rejected — 412 Precondition Failed

Method & Path	PUT /orders/1
Request Headers	Authorization: Bearer demo Content-Type: application/json If-Match: "old-version"
Request Body	<pre>{ "customerName": "Nandan", "totalAmount": 750 }</pre>
Status	412
Response Body	<pre>{ "title": "Precondition Failed", "status": 412 }</pre>

Notes: The supplied ETag does not match the current version of the resource. The server rejects the write to prevent overwriting changes made by another client — optimistic concurrency control.